MODULE 1 BODY CAVITIES

Body Cavities & Regions.

INTRODUCTION TO ANATOMY

The body cavities enclose and organize the internal organs. Knowing which cavity a structure sits in tells you what membrane wraps it, what fluid surrounds it, and what happens when that space is invaded.

COMPETENCIES W1-BODY-CAVITIES, W1-SEROUS-MEMBRANES, W1-MEDIASTINUM, W1-PERITONEAL-RELATIONSHIPS, W1-ABDOMINOPELVIC-REGIONS

BY THE END

1. Name the two major body cavities and their subdivisions, and state what each contains.
2. Define a serous membrane and name the three membrane pairs, their spaces, and their fluids.
3. Give the boundaries and contents of the mediastinum.
4. Define retroperitoneal and list the retroperitoneal structures using SAD PUCKER.
5. Draw the four quadrants and the nine regions and name the organs in each.
6. Connect an inflamed serous membrane to the condition it produces.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Week 1

drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html

The Two Major Body Cavities

A **body cavity** is a space within the body that houses and protects internal organs. There are two major cavities, and each divides further.

CAVITIES, SUBDIVISIONS, AND CONTENTS

Major cavity	Subdivision	Contents
Anterior (ventral)	Thoracic cavity	The lungs and the heart
Anterior (ventral)	Abdominopelvic cavity	The abdominal and pelvic organs
Posterior (dorsal)	Cranial cavity	The brain
Posterior (dorsal)	Spinal (vertebral) cavity	The spinal cord

The boundary. The **diaphragm** is the muscular partition that separates the thoracic cavity above from the abdominopelvic cavity below. It is the single most useful landmark in the ventral cavity, because it tells you which membranes and which fluids you are dealing with.

Serous Membranes

A **serous membrane** is a membrane that lines a cavity lacking an opening to the outside and covers the internal organs within it. Every serous membrane is a double layer folded back on itself.

- **Parietal layer**, lines the inner body wall.
- **Visceral layer**, covers the surface of an organ.

- **Serous space**, the cavity between the two layers.
- **Serous fluid**, found in that space between the visceral and parietal layers. It reduces friction so the organ can move against the wall without damage.

THE THREE NAMED SEROUS MEMBRANE PAIRS

Membrane	Wraps	Space	Fluid
Pericardium	The heart (parietal and visceral pericardium)	Pericardial cavity	Pericardial fluid
Pleura	The lungs (parietal and visceral pleura)	Pleural cavity	Pleural fluid
Peritoneum	The abdomen (parietal and visceral peritoneum)	Peritoneal cavity	Peritoneal fluid

KEY POINT

Parietal goes with the wall. Visceral goes with the organ. The two words never change meaning, no matter which cavity you are in.

The Mediastinum

The **mediastinum** is the central compartment of the thoracic cavity, the region between the two lungs.

BOUNDARIES OF THE MEDIASTINUM

Boundary	Structure
Superior	Thoracic inlet
Inferior	Diaphragm
Anterior	Sternum
Posterior	Vertebral column
Lateral	Lungs

Contents: heart, trachea, esophagus, major blood vessels, thymus.

The Peritoneum and Retroperitoneal Structures

The **retroperitoneum** is the area posterior to the peritoneum but anterior to the spine. A structure that lies there is called **retroperitoneal**. A structure fully wrapped in visceral peritoneum and suspended within the peritoneal cavity is **intraperitoneal**. The distinction is surgical: it determines the approach a surgeon takes and how an infection or a bleed will spread.

MNEMONIC: SAD PUCKER

S Suprarenal (adrenal) glands. **A** Aorta and inferior vena cava. **D** Duodenum, second and third segments. **P** Pancreas, except the tail. **U** Ureters. **C** Colon, ascending and descending. **K** Kidneys. **E** Esophagus. **R** Rectum.

Abdominal Quadrants and Regions

The four quadrants

Two planes draw the quadrant grid: a median (midsagittal) plane and a transverse plane through the umbilicus. Quadrants are the fast clinical shorthand, the version used when someone is describing pain in a triage note.

THE FOUR QUADRANTS AND THEIR KEY ORGANS

Quadrant	Key organs
Right upper (RUQ)	Liver, gallbladder
Left upper (LUQ)	Stomach, spleen
Right lower (RLQ)	Appendix, cecum
Left lower (LLQ)	Descending colon

The nine regions

Four planes, two vertical and two horizontal, divide the abdominopelvic cavity into nine regions. Regions are the more precise anatomical version, the one used when the location has to be described exactly.

THE NINE ABDOMINOPELVIC REGIONS

Region	Key structure
Right hypochondriac	Liver, gallbladder
Epigastric	Stomach
Left hypochondriac	Spleen
Right lumbar	Ascending colon
Umbilical	Small intestine
Left lumbar	Descending colon
Right iliac	Cecum, appendix
Hypogastric	Urinary bladder
Left iliac	Sigmoid colon

Inflammatory Disorders of the Serous Membranes

Each named membrane has a matching inflammatory condition. If you know the membrane and its location, you can predict the presentation.

FOUR CONDITIONS, BY MEMBRANE

Condition	Causes	Signs and symptoms	Treatment
Peritonitis	Infection, trauma, ruptured appendix	Abdominal pain, fever, rigidity	Antibiotics, surgery
Pleurisy	Viral infection, pneumonia	Sharp chest pain, dry cough	Anti-inflammatory drugs
Pericarditis	Viral infection, autoimmune disorders	Chest pain, pericardial friction rub	NSAIDs, corticosteroids
Meningitis	Bacterial or viral infection	Stiff neck, fever, headache	Antibiotics for bacterial, supportive care for viral

THE CAVITY TELLS YOU THE CONSEQUENCES

A serous cavity is a closed space, so anything that enters it has nowhere to drain. A ruptured appendix spills bowel contents into the peritoneal cavity, and the result is peritonitis across the whole abdomen rather than a local problem. A puncture of the pleural cavity lets air in, the negative pressure that holds the lung expanded is lost, and the lung collapses. Reduced serous fluid in any of the three spaces means the two layers rub instead of glide, which is why a friction rub is audible in pericarditis. Meningitis is the same logic applied to the dorsal cavity: the meninges enclose the brain and spinal cord, so an infection there raises pressure inside a rigid skull.